

WASP-3b

R-1178

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-3_160604 · created 2026-07-31T04:09:43+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457543.8708 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.105 +/- 0.013
Transit depth (Rp/Rs)^2	1.11 +/- 0.27 [%]
Orbital Inclination (inc)	88.76 +/- 1.71
Airmass coefficient 1 (a1)	1.462 +/- 0.012
Airmass coefficient 2 (a2)	-0.339 +/- 0.017
Scatter in the residuals of the lightcurve fit is	0.84 %
Best Comparison Star	None
Optimal Aperture	6.7
Optimal Annulus	11.16
Transit Duration (day)	0.1285 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.105 $\pm$ 0.013 vs 0.10538 (deviation 0.02 σ)
Duration fit vs published	0.1285 d vs 0.1167 d (deviation 3.25 σ)
Mid-time O-C	-5.6 min
Depth SNR	6.7
Residual scatter	0.84 %

Light curve

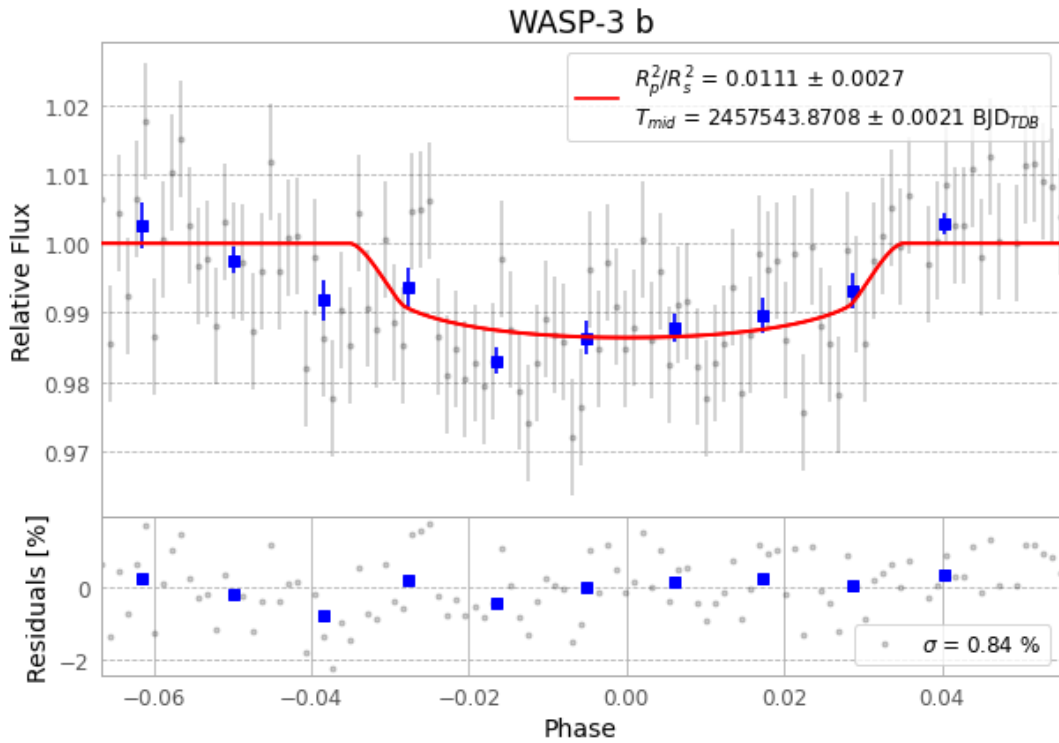


Figure 1 — FinalLightCurve_WASP-3 b_2016-06-03.png (SHA-256 46f0bda6a989bf3b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [275, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 821809e1d0e412a0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

109 FITS frames (72 MB), WASP-3160604055112.FITS → WASP-3160604111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-160604.log (3f1b6bc07cfd...), FinalLightCurve_WASP-3 b_2016-06-03.png (46f0bda6a989...), FinalLightCurve_WASP-3 b_2016-06-03.csv (a94909c3112b...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 04:09Z.